

BWC Adaptive Selector

Pair-Aware Runtime Prefetcher Selection in ChampSim

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Select locally. Coordinate globally.

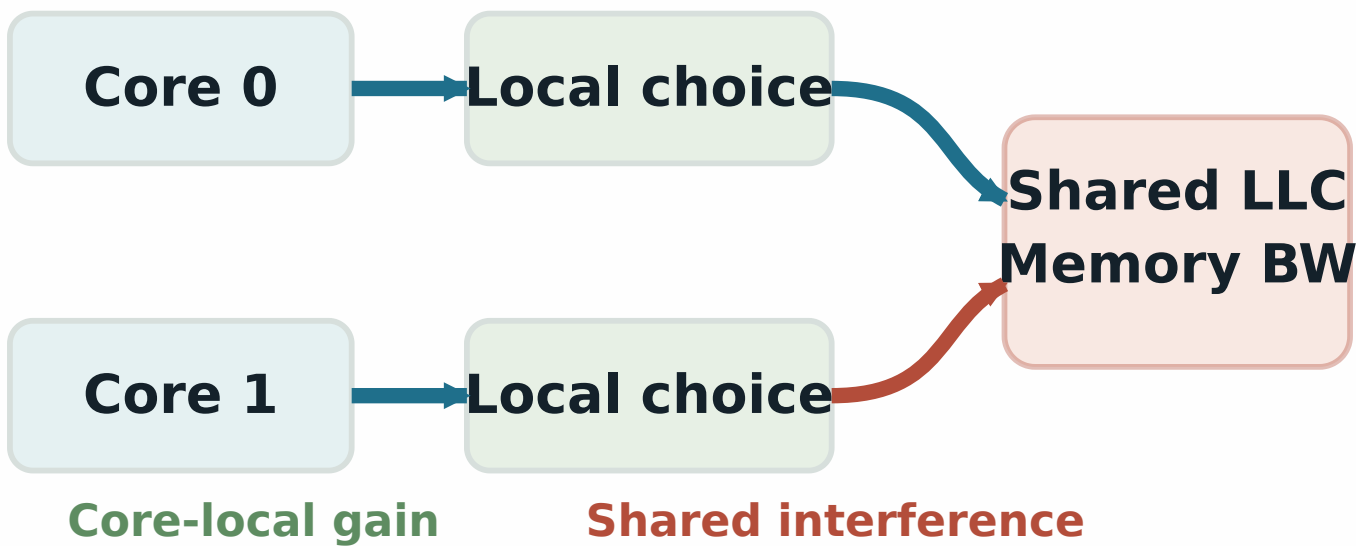
Runtime Expert Selection

Pair-Aware Coordination

Reported Gains: 4/4 Single-Core · 10/10 Pairs

Motivation: Locally Good Can Be Globally Harmful

A locally beneficial prefetcher can be globally harmful. It may hide latency for one core while polluting shared cache or consuming bandwidth for its peer.



Key Idea: 3-Step Selector Concept

BWC chooses an expert for the local stream, then checks peer risk before activation.

- Sense** — Track local stream features.
- Select** — Pick orig, FDP, or BOP.
- Coordinate** — Check peer signals before activation.

Evaluation Setup

We compare BWC against SPP_orig in ChampSim using IPC and weighted speedup gain.

SIMULATOR	BASELINE
ChampSim	SPP_orig
SINGLE-CORE	TWO-CORE
IPC gain	Weighted speedup gain

Evaluated workloads

astar mcf lbm bzip2

Reported pair groups

6 heterogeneous pairs 4 homogeneous pairs

Implementation Map

The poster maps directly to the selector stages implemented in the ChampSim prefetcher.

Feature extraction

```
compute_window_features()
page_growth · small_delta_ratio · line_growth · rfo_share
```

Local selection

```
classify_window()
orig / FDP / BOP candidate
```

Shared context

```
refresh_shared_pressure()
peer_page_growth · peer_small_delta_ratio · peer_pressure · peer_lbm_like
```

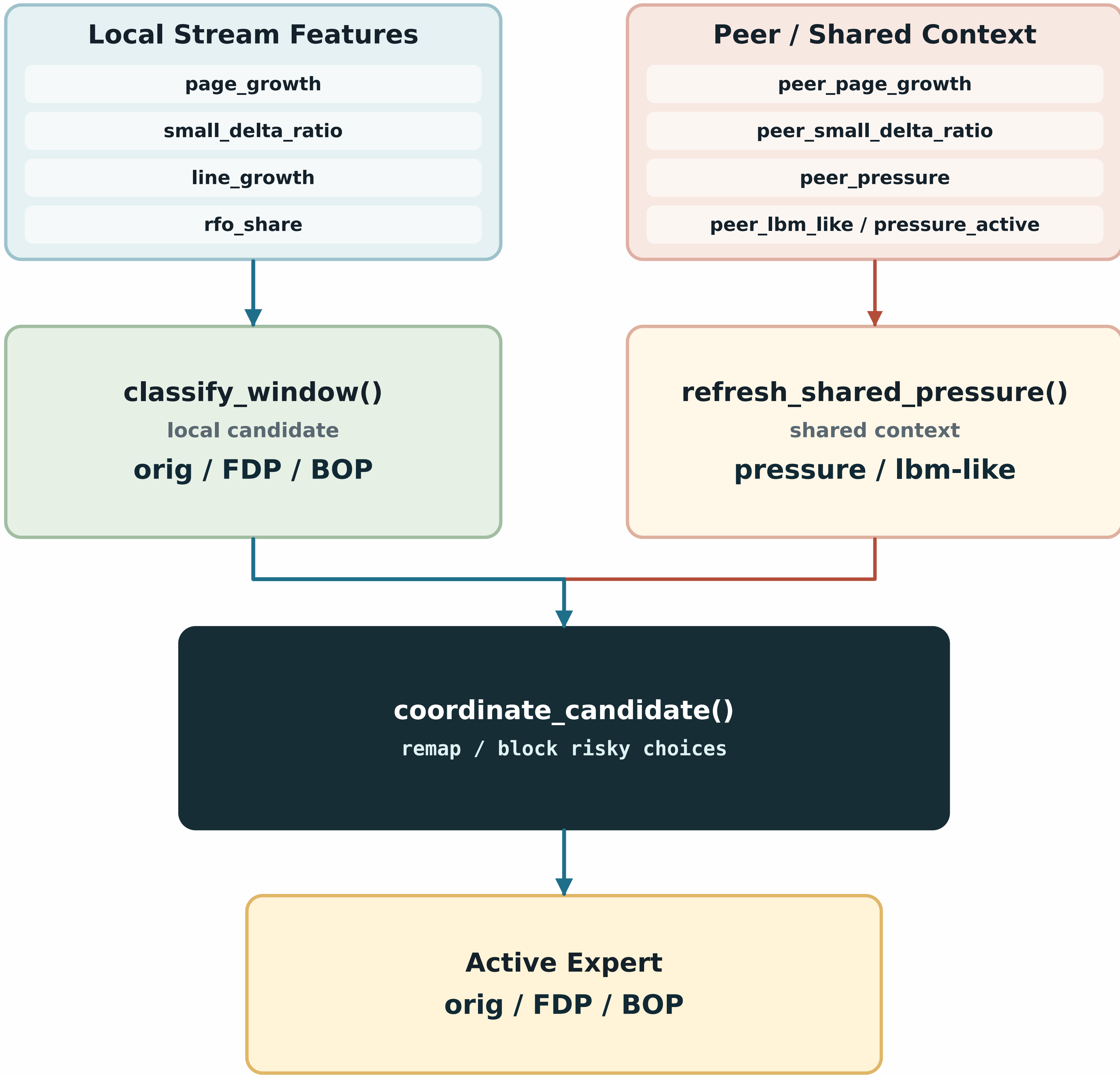
Pair coordination

```
coordinate_candidate()
remap / block risky pair choices
```

BWC Design: Select Locally, Coordinate Globally

Local stream features pick an expert; peer signals guard shared LLC and memory bandwidth.

Figure 1: BWC Adaptive Selector Pipeline



ATTACHED COORDINATOR RULES

Coordinator rule examples

RULE EXAMPLE 1
Ultra-high-page sparse pair
BOP → FDP

RULE EXAMPLE 2
Low-page dense-peer split
prefer FDP

RULE EXAMPLE 3
Pressure / lbm-like peer
block or demote BOP

Phase 2: OpenEvolve-Guided Coordinator Refinement

OpenEvolve searched coordinator-only rules. Best variant changes coordinate_candidate(), not expert implementations.

Phase 1 selector	OpenEvolve search	Phase 2 coordinator rules
bzip2 + lbm -0.03% → +0.08%	mcf + lbm -1.02% → -0.54%	Mean gain -0.224% → -0.029%
combined score -6.27 → -3.01	What changed? dense-peer split pair-scope BOP demotion lbm-like peer protection two runtime-signal guards	

Targets harmful lbm-related pair cases while leaving single-core lbm unchanged.

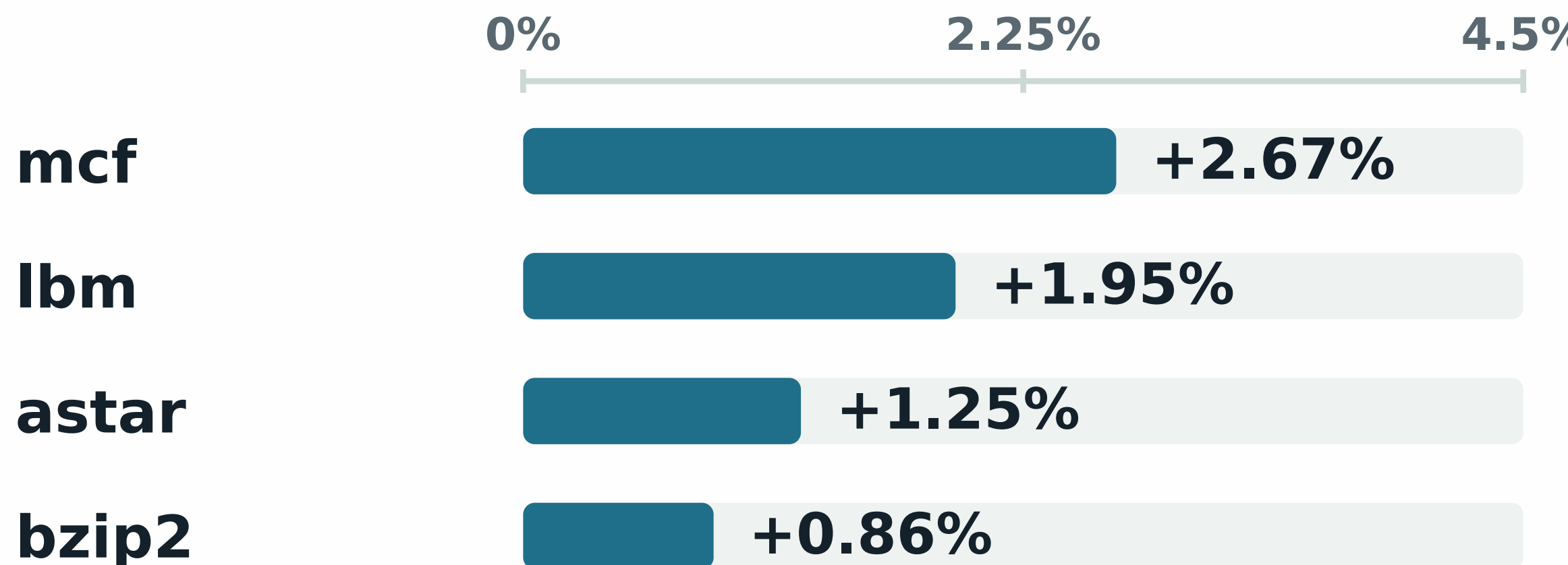
Results: Positive Gains Across Reported Summaries

Reported evaluation shows gains for every listed single-core workload and every listed two-core pair.

Max observed two-core WS gain
+4.21%
astar + astar

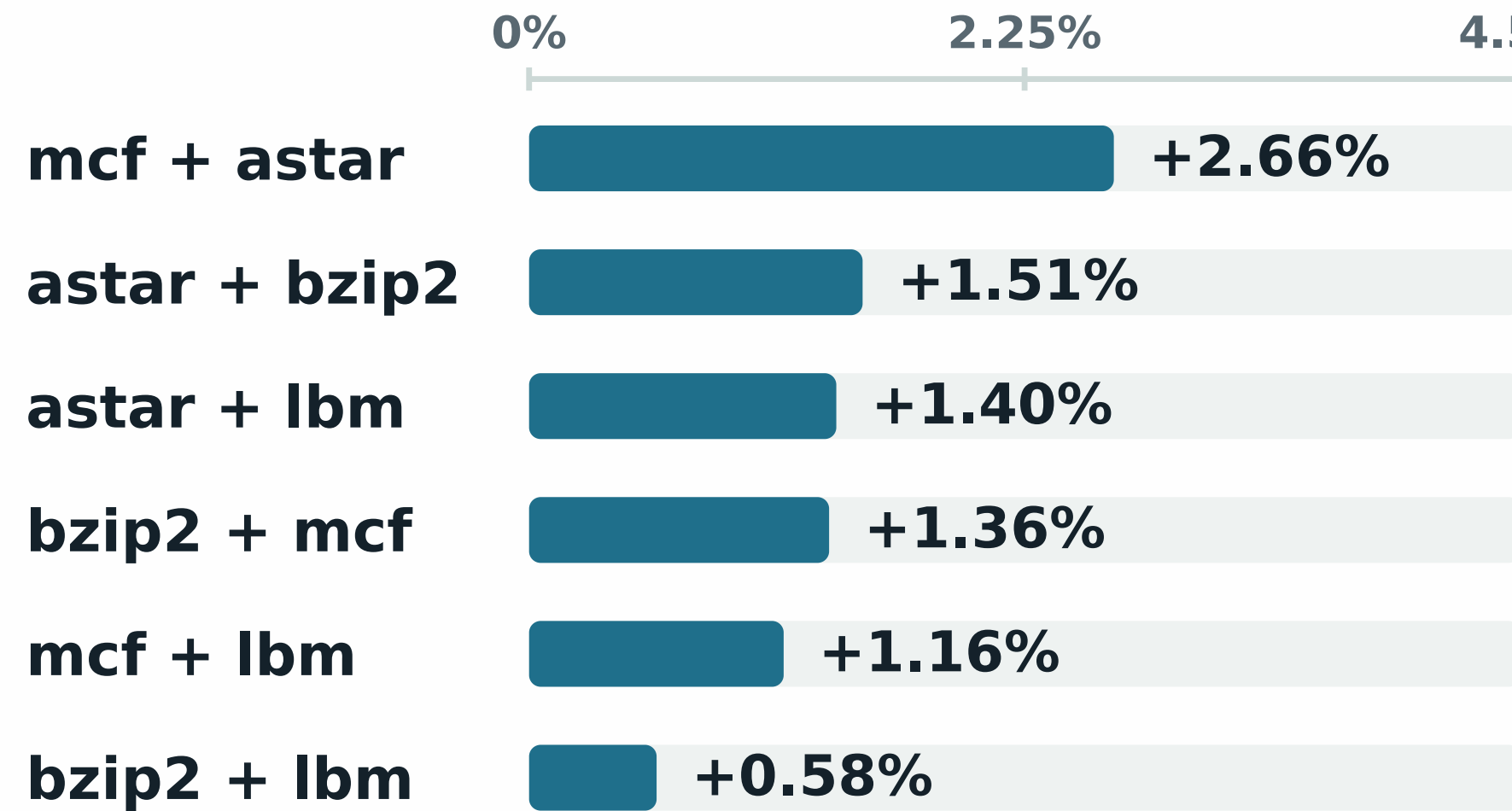
4/4 single-core ↑ 6/6 heterogeneous pairs ↑ 4/4 homogeneous pairs ↑

Single-Core IPC Gain



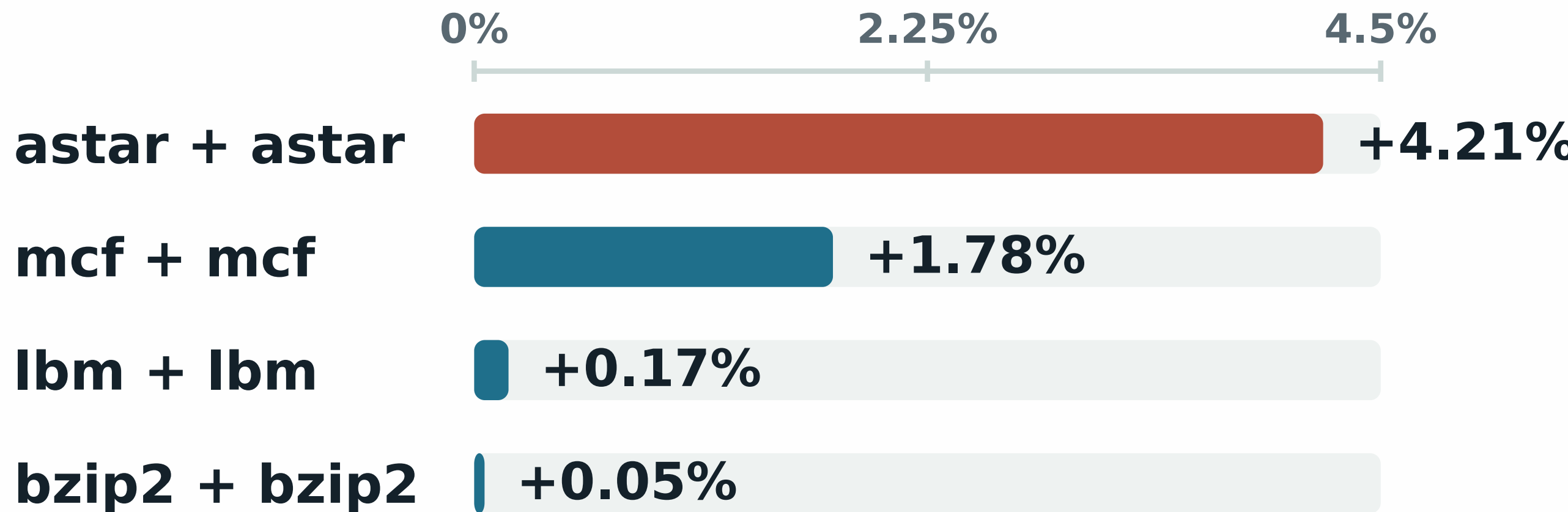
4/4 reported workloads improve.

Two-Core Heterogeneous WS Gain



6/6 mixed pairs improve.

Two-Core Homogeneous WS Gain



4/4 same-workload pairs improve.

Takeaways

- Workload-aware: runtime signals select the expert for the current stream.
- Pair-aware: peer signals guard shared LLC and memory bandwidth.
- Practical insight: prefetcher selection should coordinate across cores.